

CAMPO GRANDE, Brazil

WMO: 836120

Lat: 20.469S

Lon: 54.673W

Elev: 1833

StdP: 13.75

Time Zone: -4.00 (SAC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	46.4	49.9	32.0	28.3	61.6	36.0	33.2	67.5	22.6	73.6	20.9	72.2	10.0	130	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	16.7	95.1	68.4	93.1	69.1	91.4	69.6	76.2	85.8	75.4	85.2	74.7	84.5	12.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.6	134.0	79.9	73.0	131.2	79.6	71.9	126.3	78.8	41.0	85.9	40.3	85.3	39.7	84.8	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	19.0	17.3	DB	41.8	97.7	2.6	2.7	39.9	99.6	38.4	101.2	36.9	102.7	35.0	104.7
			WB	38.4	77.6	3.7	1.0	35.8	78.3	33.6	78.9	31.6	79.4	28.9	80.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.2	78.0	78.2	77.5	75.7	71.0	69.5	69.9	73.2	76.2	77.8	77.8	78.4
	DBStd	6.32	2.78	2.50	2.98	4.82	6.22	5.86	7.31	7.91	7.52	5.48	3.93	3.20
	HDD50	3	0	0	0	0	0	0	3	0	0	0	0	0
	HDD65	156	0	0	0	5	26	32	46	32	13	2	0	0
	CDD50	9216	869	790	851	770	652	584	618	719	787	861	835	880
	CDD65	3893	404	370	386	324	214	166	197	286	349	397	385	415
	CDH74	31951	2942	2650	2864	2488	1395	1107	1579	2764	3683	3796	3337	3346
	CDH80	12454	1058	947	1020	903	404	252	500	1267	1815	1703	1317	1268
Wind	WSAvg	8.9	8.1	7.8	7.5	7.9	8.4	9.1	9.7	10.6	10.4	9.6	8.9	8.3
Precipitation	PrecAvg	59.0	8.8	6.6	5.7	3.9	4.4	1.8	1.7	1.3	3.4	5.4	6.3	8.3
	PrecMax	75.3	20.0	12.4	10.7	7.6	10.6	8.0	7.2	4.8	9.9	11.4	11.8	14.4
	PrecMin	34.2	2.9	1.9	2.2	0.9	0.3	0.0	0.0	0.0	0.3	0.3	0.8	1.7
	PrecStd	10.2	4.1	2.5	2.3	2.0	3.0	1.9	1.7	1.5	2.5	2.6	2.8	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.9	91.8	91.9	91.5	88.2	86.4	89.4	94.6	98.4	98.3	93.5	93.2
		MCWB	72.7	72.4	72.7	72.0	69.4	66.1	64.7	65.7	66.5	69.6	72.1	71.5
	2%	DB	90.0	89.8	89.8	89.4	86.2	84.5	87.4	91.7	95.3	94.9	91.5	91.2
		MCWB	73.0	72.7	73.1	71.1	69.1	65.9	63.9	64.7	66.6	70.9	71.5	72.3
	5%	DB	88.0	87.9	87.9	87.7	84.4	82.8	85.1	89.8	93.2	91.7	89.6	89.3
		MCWB	72.8	72.6	72.5	70.3	68.4	65.6	64.1	64.7	66.5	71.2	71.4	72.6
	10%	DB	86.0	86.0	86.0	85.8	82.4	80.8	82.8	87.7	91.1	89.4	87.6	87.4
		MCWB	72.4	71.8	71.7	69.7	67.7	65.6	63.4	64.1	66.4	71.3	71.2	72.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.6	76.4	77.2	75.3	74.5	72.6	69.5	70.6	73.6	75.8	76.3	76.9
		MCDB	86.2	85.8	86.3	85.1	84.1	80.3	79.9	86.3	88.8	88.4	86.0	85.8
	2%	WB	75.5	75.4	75.7	74.2	72.8	70.4	68.0	68.5	72.1	74.4	75.0	75.6
		MCDB	84.9	85.1	84.9	84.2	81.2	77.9	79.3	84.3	86.8	87.5	85.5	85.0
	5%	WB	74.7	74.5	74.7	73.3	70.9	68.8	66.7	67.1	70.8	73.4	74.1	74.7
		MCDB	83.8	84.0	83.7	83.2	78.3	77.1	78.6	83.1	84.7	86.2	84.4	84.1
	10%	WB	73.8	73.7	73.7	72.2	69.6	67.3	65.4	65.6	69.2	72.4	73.2	74.0
		MCDB	82.7	82.7	82.4	81.0	77.1	76.4	77.7	81.6	82.2	84.3	83.3	83.1
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.1	16.7	17.6	17.3	18.4	20.5	21.9	21.4	18.9	18.0	16.7
		MCDBR	19.2	18.8	19.0	19.4	19.3	20.3	22.8	23.3	24.0	22.0	20.2	19.3
		MCWBR	7.0	6.5	6.9	6.7	7.4	7.8	8.3	8.4	7.9	7.4	7.5	7.1
	5% WB	MCDBR	16.5	16.3	16.6	16.5	16.1	17.3	19.8	21.6	20.6	18.7	17.2	16.5
		MCWBR	7.0	6.7	6.9	7.0	7.3	8.1	8.8	9.1	8.4	7.4	7.2	7.1
Clear-Sky Solar Irradiance	taub	0.392	0.398	0.385	0.365	0.349	0.373	0.370	0.433	0.574	0.490	0.409	0.404	
	taud	2.458	2.439	2.470	2.509	2.506	2.437	2.417	2.190	1.825	2.116	2.361	2.405	
	Ebn at Noon	302	297	293	286	279	264	269	260	236	267	295	299	
	Edh at Noon	38	38	36	32	30	31	32	43	67	52	42	40	
All-Sky Solar Radiation	RadAvg	1860	1842	1773	1602	1296	1204	1297	1525	1608	1763	1895	1903	
	RadStd	186	100	126	103	113	88	108	97	144	103	145	131	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page